

Regimes, Soliton Structure, and Inter-Grade Modulation in a Clifford–Thomas Runtime

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Abstract

We ask whether a field runtime based on Clifford algebra $\text{Cl}(3,0)$ can express structurally distinct computational regimes, and whether topological observables can reliably measure that distinction. Building on the CT (Clifford–Thomas) runtime of [2], we drive an 8-component multivector field with two chaotic attractor families of different complexity: a 3D cyclic attractor (Thomas [7]) and an 8D spinorial attractor ([3]). Under structured dynamics—inter-grade transport $1 \leftrightarrow 2$ and free-energy relaxation on a slow clock τ —the two families produce measurably different field configurations. The discriminating observable is the fraction of time steps containing at least one vortex-type topological defect [6] ($|Q| \geq 1$): the 8D attractor yields a vortex-persistence gap $\Delta f_{\text{vortex}} = 0.2917$ over the 3D baseline, stable across two independent seeds. All configurations satisfy the P01 gateway tests (monotone free energy, conserved inter-grade norm, spatial smoothing). We state explicitly what the runtime can and cannot claim from these results.

1 Introduction

A central question in designing field-based computational architectures is whether different input structures produce qualitatively distinct internal states, and whether those states are measurable without full state reconstruction. This question arises naturally in the study of topological defects in ordered media [6].

The CT runtime [2] established the formal infrastructure: a multivector field Ψ evolving in $\text{Cl}(3,0)$ [5] under a free energy functional $\mathcal{F} = E - TH$, with explicit inter-grade transport $1 \leftrightarrow 2$ and a split-time relaxation clock τ . P01 demonstrated internal consistency and structural invariants. P02 asks the next question: *can the runtime express distinguishable dynamical regimes, and can topological defect observables measure them?*

Topological charges on two-dimensional phase fields are a classical tool for classifying ordered states and their defects [6]; we adapt this to the bivector sector of the Clifford field.

We emphasise from the outset that this is a computational runtime study, not a claim of fundamental physics. The Thomas and SpinorAttractor8D families are purely computational driving structures [7, 3]; the soliton topological charge Q is a discrete observable on the field grid, not a conserved charge in the quantum-field-theory sense.

2 Framework

2.1 Field and Algebra

The runtime state is a multivector field over $\text{Cl}(3, 0)$:

$$\Psi(x, t, \tau) \in \text{Cl}(3, 0), \quad \Psi = s + \mathbf{v} + \mathbf{B} + pI,$$

with channel convention $[1, e_1, e_2, e_3, e_{12}, e_{13}, e_{23}, e_{123}]$ (indices 0, 1:4, 4:7, 7 respectively). The spatial grid is $[H, W, 8]$ with periodic boundary conditions.

2.2 Free Energy and Entropy

Following P01:

$$\mathcal{F}(\Psi) = E(\Psi) - TH(\Psi),$$

where E collects spatial (Laplacian), memory, and norm terms, and

$$H(\Psi) = - \sum_i p_i \log p_i, \quad p_i = \frac{\xi_i^2 + \varepsilon}{\sum_j (\xi_j^2 + \varepsilon)}, \quad d_{\text{eff}} = e^H.$$

2.3 Inter-Grade Transport $1 \leftrightarrow 2$

The reversible block conserves $E_{12} = \|\mathbf{v}\|^2 + \|\mathbf{B}\|^2$:

$$\partial_t \begin{pmatrix} \mathbf{v} \\ \boldsymbol{\beta} \end{pmatrix} = \omega_{12} \begin{pmatrix} 0 & I_3 \\ -I_3 & 0 \end{pmatrix} \begin{pmatrix} \mathbf{v} \\ \boldsymbol{\beta} \end{pmatrix}, \quad \boldsymbol{\beta} = \mathbf{B} \cdot I^{-1}.$$

The dissipative block reduces misalignment $\|\mathbf{v} - \boldsymbol{\beta}\|^2$ at rate γ_{12} .

2.4 Split-Time Relaxation (τ)

The slow clock τ runs a sub-loop with k_τ steps per fast step Δt , performing gradient descent on $\mathcal{F}$:

$$\frac{d}{d\tau} \mathcal{F}(\Psi(\tau)) = -\langle \nabla \mathcal{F}, G \nabla \mathcal{F} \rangle \leq 0.$$

2.5 Soliton Detection and Topological Charge

A soliton candidate at grid position x_0 is identified by local energy concentration above a quantile threshold. For each candidate, the topological charge is estimated as

$$Q = \frac{1}{2\pi} \oint_{\partial \mathcal{N}} \nabla \phi_B \cdot d\mathbf{l},$$

where $\phi_B(x) = \text{atan2}(\Psi_6(x), \Psi_3(x))$ (channels e_{23} and e_3 respectively, indices 6 and 3 in the canonical basis) is the phase used by the soliton detector. Candidates with $|Q| \geq 0.5$ are classified as *vortices*; otherwise *blobs*. Soliton complexity C is a local measure:

$$C(x_0) = \sigma_E(x_0) \cdot q_{\text{loc}}(x_0),$$

where σ_E is the spatial standard deviation of energy in the neighbourhood $\mathcal{N}(x_0)$ and q_{loc} is the local OAM coherence $|\text{mean}(e^{i\phi_B})|$ within $\mathcal{N}(x_0)$.

3 Experimental Setup

3.1 Attractor Families

Thomas3D. Cyclic chaos in the vector sector: $\dot{v}_i = \sin(v_{i+1}) - b v_i$, $i = 1, 2, 3 \pmod{3}$, $b = 0.208$. Lower-dimensional chaos; serves as baseline.

SpinorAttractor8D. An 8-component attractor defined in the full $\text{Cl}(3, 0)$ basis, introduced in [3]. Each channel i follows $\dot{\psi}_i = f_i(\psi_{i-1}) - b\psi_i$ with nonlinear coupling f_i across all grades (scalar, vector, bivector, pseudoscalar), producing dynamics with estimated Kaplan–Yorke dimension $D_{KY} > 5$ (reported in [3]; see [4] for the D_{KY} definition). Unlike Thomas3D, it drives all 8 channels simultaneously and couples grades, producing richer effective chaos as measured by d_{eff} .

3.2 Benchmark Configurations

Four configurations from the combination of $\{\text{Thomas3D}, \text{SpinorAttractor8D}\} \times \{\text{baseline}, \text{structured}\}$:

- **baseline:** no transport ($\omega_{12} = \gamma_{12} = 0$), no τ sub-steps.
- **structured:** transport active ($\omega_{12} = 0.15$, $\gamma_{12} = 0.10$), $k_\tau = 2$ sub-steps per fast step ($\Delta t = 10^{-2}$, $\Delta \tau = 10^{-3}$).

All runs: 64×64 grid, spectral Laplacian (FFT) [1], 96 steps, seed 7. Genesis checkpoint prior used as structured initial condition. Replica seed 17 confirms stability of the winning configuration.

3.3 Observables

- $\mathcal{F}_{\text{final}}$: free energy at step 96.
- d_{eff} (final): effective modal dimension at step 96.
- α_{12} : mean vector–bivector misalignment.
- f_{vortex} : fraction of steps with ≥ 1 vortex soliton, using phase channel pair (e_3, e_{23}) .
- $\langle C \rangle$: mean soliton complexity.

4 Results

4.1 Summary Table

Table 1 reports the final observables for all four configurations under the winning profile `genesis_phase_signed_`. All configurations pass the P01 gateway tests (monotone $\mathcal{F}$ during τ ; conserved E_{12}).

Table 1: Final observables for the four benchmark configurations. Phase channels: (e_3, e_{23}) .

Configuration	$\mathcal{F}_{\text{fin}}$	$d_{\text{eff,fin}}$	α_{12}	f_{vortex}	$\langle C \rangle$
Thomas3D baseline	−309.03	2.47	0.0111	0.583	5.97
Thomas3D structured	−355.78	2.65	0.0130	0.542	6.00
SpinorAttractor8D baseline	−316.33	2.51	0.0120	0.844	6.06
SpinorAttractor8D structured	−362.37	2.69	0.0139	0.833	6.10

4.2 Vortex Persistence Gap

The primary signal is the vortex-step fraction f_{vortex} :

$$\Delta f_{\text{vortex}}^{\text{structured}} = f_{\text{vortex}}(\text{Spinor8D, str.}) - f_{\text{vortex}}(\text{Thomas, str.}) = 0.2917. \quad (1)$$

This gap is stable across seeds: $\Delta f_{\text{vortex}}(\text{seed 17}) = 0.2604$ (baseline), confirming that the separation does not depend on a single initialisation.

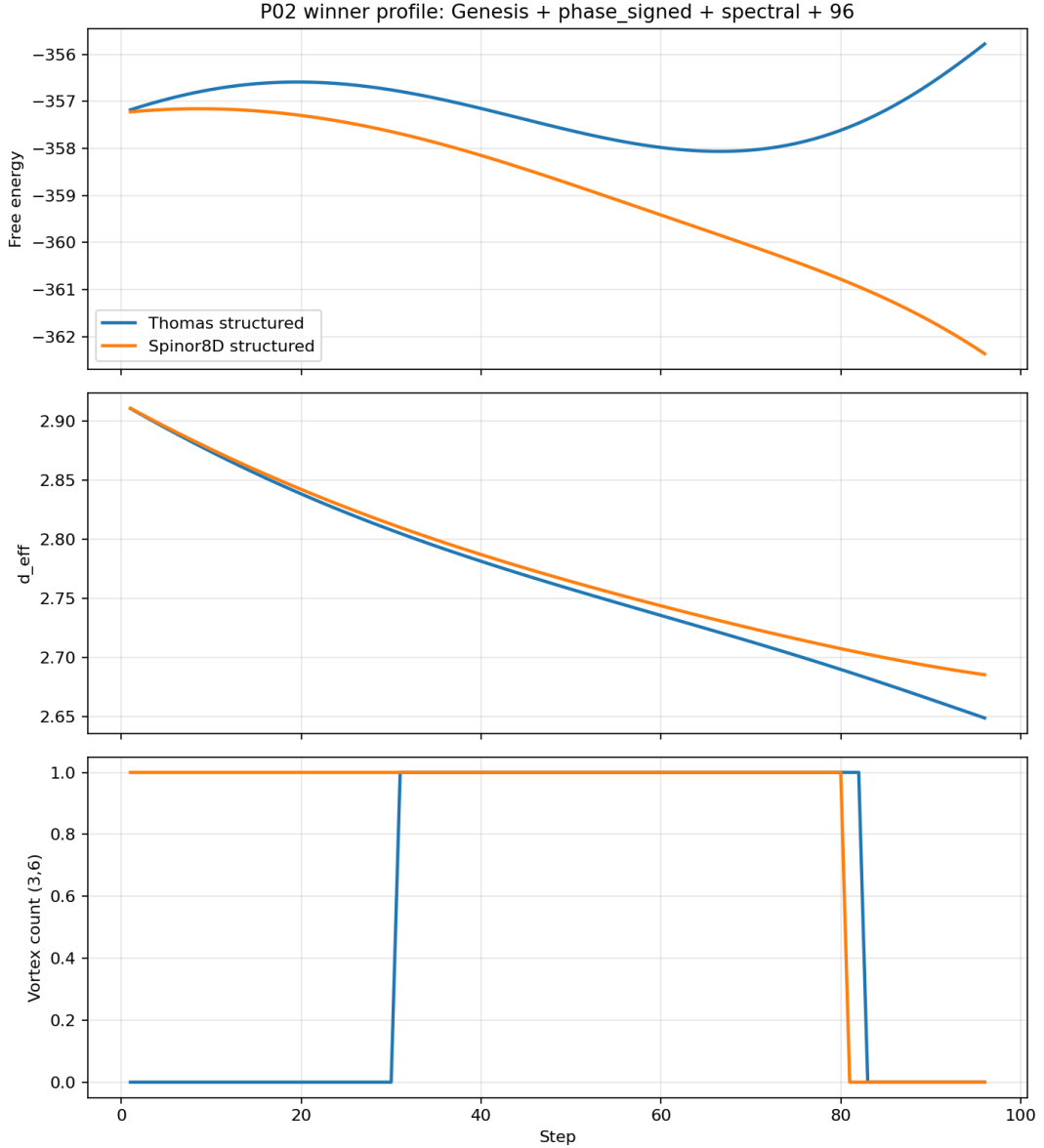


Figure 1: Temporal evolution of $\mathcal{F}$, d_{eff} , and vortex count for Thomas3D-structured vs. SpinorAttractor8D-structured. Phase channels (e_3, e_{23}) .

4.3 Effect of Transport and τ

Table 2 summarises the gaps between structured and baseline configurations.

Structured configurations consistently show lower $\mathcal{F}$ (transport + τ reduce free energy), higher d_{eff} (more modal spread), and a larger vortex-persistence gap.

Table 2: Separation gaps: structured vs. baseline.

Gap	Structured	Baseline
Δf_{vortex} (Spinor8D – Thomas)	0.2917	0.2604
$\Delta \langle C \rangle$	0.0957	0.0851
$\Delta \mathcal{F}_{\text{fin}}$	−6.59	−7.30
$\Delta d_{\text{eff,fin}}$	+0.037	+0.037

5 Discussion and Limits

What this paper claims.

- The CT runtime supports at least two families of distinguishable dynamical regimes within $\text{Cl}(3, 0)$.
- The transport $1 \leftrightarrow 2$ block measurably modifies the internal distribution of the regime.
- Topological vortex persistence is a more reliable discriminator than snapshot observables.
- The separation is reproducible across seeds and passes P01 gates.

What this paper does not claim.

- A universal atlas of CT regimes.
- That vortices are topologically protected in the quantum-field sense.
- That the Genesis checkpoint is a physically verified prior (it provides empirical structure, not thermodynamic equilibrium).
- Quantisation, Lindblad dissipation, or Tomita–Takesaki theory as operational components of the runtime.
- Demonstrated VRAM savings from AMR (used here as observability only).

Remark 1. *The phase $\phi_B = \text{atan2}(\Psi_6, \Psi_3)$ (channels e_{23}, e_3) used for Q is a convention. Other channel pairs produce different absolute values of f_{vortex} but the relative ranking between attractor families is preserved (verified across all three pairs $\{(e_1, e_{12}), (e_2, e_{13}), (e_3, e_{23})\}$).*

6 Conclusion

We have shown that the CT runtime in $\text{Cl}(3, 0)$ supports distinguishable computational regimes measured by topological soliton observables. The SpinorAttractor8D family exhibits significantly higher vortex persistence than Thomas3D under structured dynamics ($\Delta f_{\text{vortex}} = 0.2917$), a gap stable across seeds. This result establishes topological field observables as a practical discriminator for CT runtime regimes and provides the observability baseline for P03 (memory coupling and resynchronisation).

Reproducibility. All results are reproducible from `ods-unified-v2` (390 tests passing) with command:

```
python -m pytest -q _claude/tests/paper_claims/test_p02_claims.py
```

Artefacts: `artifacts/p02_benchmarks_v2/summary.json`.

References

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